

Unlocking Path Awareness for Legacy Applications through SCION-IP Translation in eBPF

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FAKULTÄT FÜR
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The SCION Internet Architecture



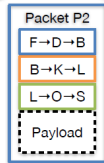
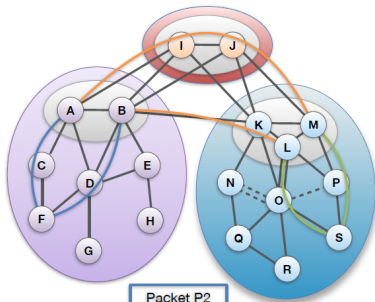
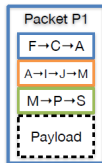
Path-based Network Architecture

Control Plane - Routing

- ❖ **Constructs** and **Disseminates** Path Segments

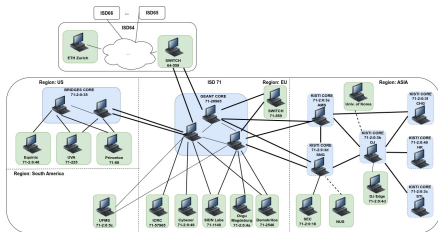
Data Plane - Packet forwarding

- ❖ **Combine** Path Segments to Path
- ❖ Packets contain Path
- ❖ Routers forward packets based on Path
 - ▶ Simple routers, stateless operation



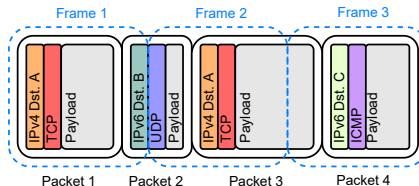
SCION Deployments

- ▶ Commercial deployments: Secure Swiss {Finance|Health} Network
- ▶ SCIARA: SCION Education, Research and Academic Network



- ▶ Few native SCION applications

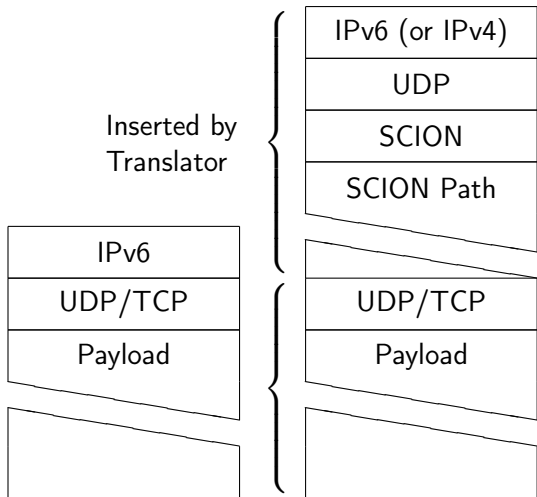
- ▶ Enabler: **SCION-IP Gateway (SIG)**
 - ▶ VPN router for IPv4/6 traffic over SCION networks
 - ▶ SIG encapsulation protocol is incompatible with native SCION applications



- ▶ No end host path control

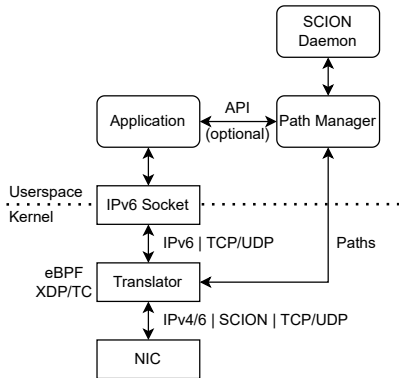
SCION-IP Translation

- ▶ Idea: Legacy IPv6 applications can address SCION hosts by **encoding SCION ISD-ASN in IP address**
- ▶ Translate packets 1:1 at boundary between applications and network
 - ▶ IP→SCION inserts SCION path
 - ▶ SCION→IP strips SCION path out
- ▶ End-host with SCION-IP translator is equivalent to native SCION host
- ▶ Legacy IP application can talk to native SCION application



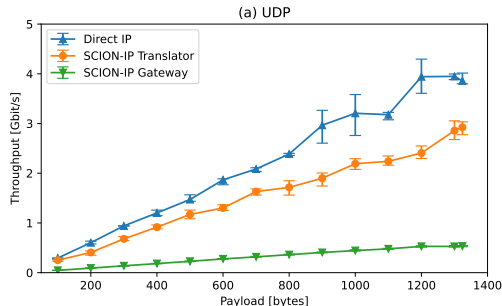
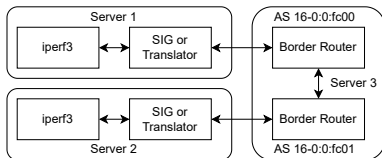
eBPF Prototype

- ▶ **XDP** on ingress; **TC** on egress
- ▶ Paths are cached in eBPF map
- ▶ Data plane requests paths via ringbuffer
- ▶ Path requests are fulfilled by SCION daemon in user space
- ▶ No changes to the application required
- ▶ Optional path control possible via system API



Evaluation

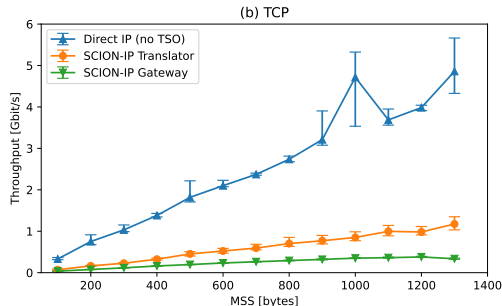
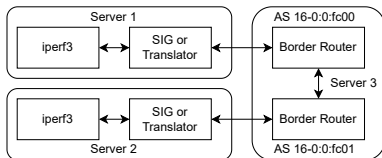
- ▶ Comparing throughput of direct IP, SCION-IP Gateway and SCION-IP Translator (eBPF) using iperf
- ▶ Performance limited by Open-source SCION stack
- ▶ Testbed with two ASes and two hosts:



- ▶ Forced TSO/GSO and GRO off to avoid corrupting packets
- ▶ Open-Source SIG limited by SCION dispatcher

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Conclusions / Outlook

- ▶ Prototype works, but performance is still limited
- ▶ Investigate Generic Segmentation Offload for tunneled traffic
- ▶ Possible implementation as kernel module
- ▶ SCION-IP Translation can replace SIG as IP gateway to SCION
- ▶ SCION-IP application interoperability enables incremental move towards native SCION
- ▶ Other SCION-IP Translators are in development: Android (Java), Windows (C++), Intel Tofino (P4)

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Full Technical Report



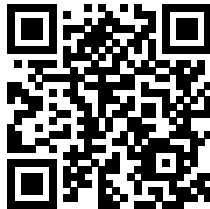
[https:
//www.netsys.ovgu.de/
netsys_media/publications/
NetSys_TR_2024_01.pdf](https://www.netsys.ovgu.de/netsys_media/publications/NetSys_TR_2024_01.pdf)

Source Code and Resources



[https:
//github.com/netsys-lab/
scion-ip-translator](https://github.com/netsys-lab/scion-ip-translator)

SCI ERA Network



[https:
//sciera.readthedocs.io](https://sciera.readthedocs.io)

SCION-Mapped IP Addresses

- ▶ SCION ISD is 16 bit, but only 12 bits used for allocations
- ▶ SCION ASN is 48 bit, encoded as 20 bit (2^{19} BGP + 2^{19} SCION ASNs)

128-bit IPv6 address					
8 bit	12 bit	20 bit	24 - m bit	m bit	64 bit
global routing prefix				subnet ID	interface ID
SCION prefix	ISD	ASN	local routing prefix	subnet ID	interface ID

- ▶ IPv4 encoding analogously to IPv4-mapped IPv6 addresses

128-bit IPv6 address					
8 bit	12 bit	20 bit	24 bit	32 bit	32 bit
global routing prefix				interface ID	
SCION prefix	ISD	ASN	0	0x0000ffff	IPv4 address